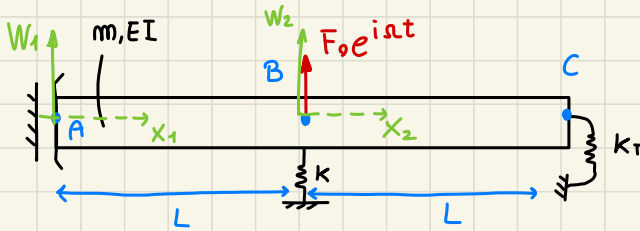


ESE 18 VIBRATION OF DPS and MODAL APPROACH

Bending Vibration of Beams

P2. 01/02/2022 Tde

(TORSIONAL SPRING on the right end) K_T



define functions to define constrain of displacement $\Rightarrow$ DISCONTINUITY on the middle

W_1, W_2 respect the two beam portion general expression W_k :

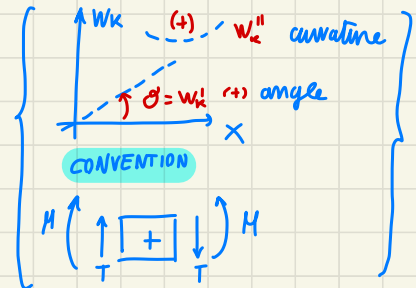
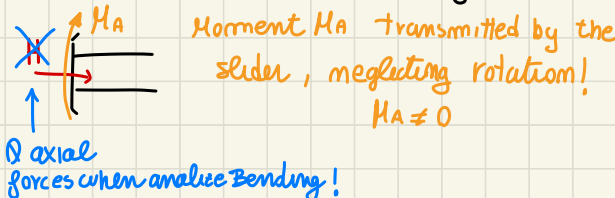
$$\begin{cases} W_k(x,t) = [A_k \cos(\gamma x) + B_k \sin(\gamma x) + C_k \cosh(\gamma x) + D_k \sinh(\gamma x)] \cos(\omega t + \varphi) \\ \frac{\partial W_k}{\partial x}(x,t) = \gamma [-A_k \sin(\gamma x) + B_k \cos(\gamma x) + C_k \sinh(\gamma x) + D_k \cosh(\gamma x)] \cos(\omega t + \varphi) \\ \frac{\partial^2 W_k}{\partial x^2}(x,t) = \gamma^2 [-A_k \cos(\gamma x) - B_k \sin(\gamma x) + C_k \cosh(\gamma x) + D_k \sinh(\gamma x)] \cos(\omega t + \varphi) \\ \frac{\partial^3 W_k}{\partial x^3}(x,t) = \gamma^3 [A_k \sin(\gamma x) - B_k \cos(\gamma x) + C_k \sinh(\gamma x) + D_k \cosh(\gamma x)] \cos(\omega t + \varphi) \end{cases} \quad (k=1,2)$$

We have 2 funct. and 8 unknown A_k, B_k, C_k, D_k $k=1,2 \Rightarrow 8$ conditions!
for the extremity where the discontinuity acts

@ $x_1=0$: $W_1(0)$? unkwn for slider... ALL CONDIT. Valid $\forall t$
SLIDER @ $x_1=0$ having a SLIDER
 $\nparallel$ so it cannot ROTATE!

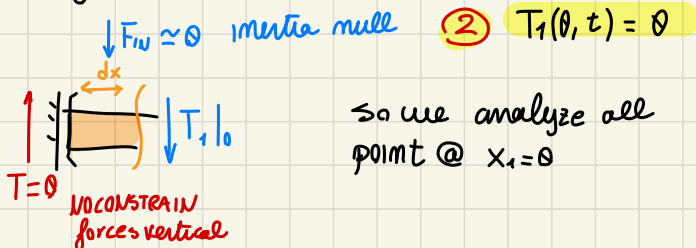
① $\theta_1(0,t) = 0 \quad \forall t$

② try to write condition respect Bending moment
In section A which is $\neq 0$ because of slider

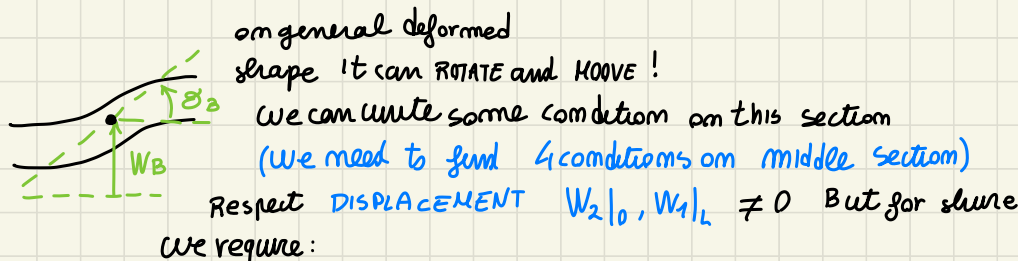


$$\begin{cases} M_k(x,t) = EI \frac{\partial^2 W_k}{\partial x^2}(x,t) \\ T_k(x,t) = EI \frac{\partial^3 W_k}{\partial x^3}(x,t) \\ \gamma^4 = \omega^2 m / EI \end{cases}$$

$M_A \neq 0 \rightarrow$ so the dependency on w'' is hard...
consider the shear force



• on section $x_1 = L, x_2 = 0$



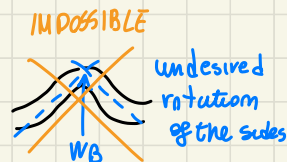
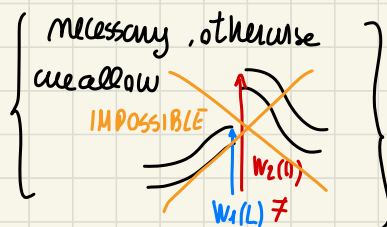
(3) $w_B = w_1|_L = w_2|_0$ unknown But forcing constraint CONTINUITY

$\forall t \quad w_1(L, t) - w_2(0, t) = 0$

(4) by an analysis on w_k , so the rotation θ_B unknown, But we want also orientation continuity between the beam parts!

$\theta_B = \theta_1|_L = \theta_2|_0 \Rightarrow \theta_1(L, t) - \theta_2(0, t) = 0$

$\frac{\partial w_1}{\partial x_1}|_L - \frac{\partial w_2}{\partial x_2}|_0 = 0$



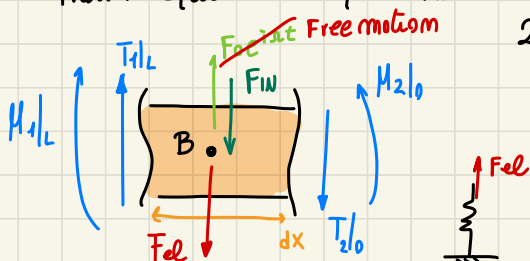
avoid discontinuity of ROTATION!

↓ possible on STRINGS, where $EI = 0$ so bending stiffness null

NOT here! $\rightarrow EI \neq 0!$

section (B), others eq.

from equilibrium equations:



2 eq of equilibrium:

$\sum F_w = 0$

$\sum M_B = 0$

Dynamic equilibrium

↳

$$\sum M_B = 0 \Rightarrow M_1|_L + T_1|_L \frac{dx}{2} + T_2|_0 \frac{dx}{2} - M_2|_0 = 0$$

$dx \approx 0$ negligible

$$M_1(L, t) - M_2(0, t)$$

$$(5) \quad \left. \frac{\partial^2 W_1}{\partial x_1^2} \right|_L - \left. \frac{\partial^2 W_2}{\partial x_2^2} \right|_0 = 0$$

$$\sum F_v = 0 \quad T_1|_L - T_2|_0 - F_{im} - F_{el} = 0$$

$$T_1(L, t) - \left. \frac{\partial T_1}{\partial x_1} \right|_L \frac{dx}{2} - \left(T_2(0, t) + \left. \frac{\partial T_2}{\partial x_2} \right|_0 \frac{dx}{2} \right) - mdx \left. \frac{\partial^2 W_1(x, t)}{\partial t^2} \right|_0 - KW_B = 0$$

$dx \approx 0$ negligible

$$(6) \quad T_1(L, t) - T_2(0, t) - K(W_2(0, t))' = 0$$

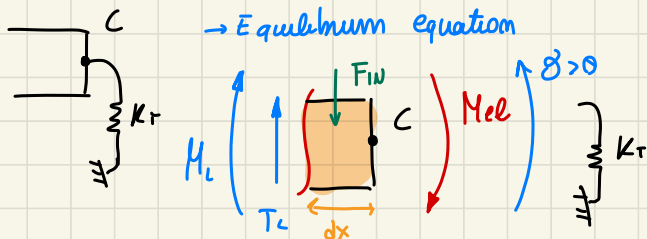
$\text{we use } W_2 = W_B \text{ so we evaluate it @ } x_2 = 0, \text{ simpler}$

$$EI \left. \frac{\partial^3 W_1}{\partial x_1^3} \right|_L - EI \left. \frac{\partial^3 W_2}{\partial x_2^3} \right|_0 - KW_2(0, t) = 0$$

- Respect the final section C

$W_C \neq 0 \quad W'_C \neq 0$ can move & rotate

→ Equilibrium equation



TORSIONAL STIFFNESS

$$K_T \cdot \vartheta = M_{el}$$

ELASTIC MOMENT

M_{el} direction COUNTERCLOCKWISE

because: for our convention of rotation $\vartheta > 0$ if COUNTERCLOCK

so M_{el} physically is opposite

because ELASTIC moment has $W > 0$ positive work

$K_T \uparrow$ reduce the displacem. same for K_T

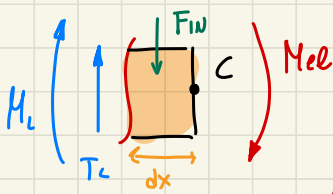
TORSIONAL $K_T \uparrow$ lead to a behaviour like slider, NO ROTATION!

elastic force perform a negative WORK

OR means that it

introduce energy! UNSTABLE effect!

$K_T \uparrow$ means higher energy introduced!
 $\vartheta \uparrow$ WRONG!



$$\sum M_c = 0$$

$$M_2|_L + \cancel{T_2|_L dx} - \cancel{F_N \frac{dx}{2}} + M_{el} = 0$$

mögliche

$$(7) \quad M_2(L, t) - \frac{\partial M_2}{\partial x}|_L dx + K_T \cdot \delta_2(L, t) = 0$$

(8) equilibrium transversal direction

$$\sum F_w = 0 \quad T_L - F_N = 0$$

$$T_2(L, t) - \frac{\partial T_2}{\partial x}|_L dx - m dx \frac{\partial^2 w_2}{\partial t^2} = 0$$

↓ we have found all 8 CONSTRAINTS: to write as equation

$$(1) \quad \gamma [B_1 + D_1] \cos(\omega t + \varphi) = 0$$

$$(2) \quad \gamma^3 [-B_1 + D_1] \cos(\omega t + \varphi) = 0$$

$$(3) \quad [A_1 \cos(\delta L) + B_1 \sin(\delta L) + C_1 \cosh(\delta L) + D_1 \sinh(\delta L)] - [A_2 + C_2] = 0$$

$$(4) \quad [-A_1 \sin(\delta L) + B_1 \cos(\delta L) + C_1 \sinh(\delta L) + D_1 \cosh(\delta L)] - [B_2 + D_2] = 0$$

$$(5) \quad [-A_1 \cos(\delta L) - B_1 \sin(\delta L) + C_1 \cosh(\delta L) + D_1 \sinh(\delta L)] - [-A_2 + C_2] = 0$$

$$(6) \quad \gamma^3 EI [A_1 \sin(\delta L) - B_1 \cos(\delta L) + C_1 \sinh(\delta L) + D_1 \cosh(\delta L)] - \gamma^3 EI [-B_2 + D_2] - K [A_2 + C_2] = 0$$

$$(7) \quad \gamma^2 EI [-A_2 \cos(\delta L) - B_2 \sin(\delta L) + C_2 \cosh(\delta L) + D_2 \sinh(\delta L)] + K_T \gamma [-A_2 \sin(\delta L) + B_2 \cos(\delta L) + C_2 \sinh(\delta L) + D_2 \cosh(\delta L)] = 0$$

$$(8) \quad A_2 \sin(\delta L) - B_2 \cos(\delta L) + C_2 \sinh(\delta L) + D_2 \cosh(\delta L) = 0$$

$$\underline{x} = \{A, B, C, D, A_2, B_2, C_2, D_2\}^T$$

$\omega_1, \omega_2, \dots$
 $\uparrow$
 such that $\delta \det$
 $\uparrow \delta_1, \delta_2, \dots$
 $\det[H(\delta)]$

↳ we write

$$[H(\delta)]_{8 \times 8} \Rightarrow [H(\delta)] \underline{x} = 0$$

from [H] we answer Q1, Q2!

Boundaries \ Natural freq $\Rightarrow$ Write Hom
MATLAB + RUN

Q3) FRF

respect $F_0 e^{i\omega t}$ INPUT through software code we compute

$\bar{m}, \bar{k}, \bar{c}$ due to the BEAM! $\forall$ mode of VIBRATION

BUT $\Rightarrow$ because there are $\begin{cases} \text{LINEAR SPRING } K \\ \text{TORSIONAL SPRING } K_T \end{cases} \Rightarrow$ to add on modal stiffness!

and if we had

additional mass $\Rightarrow$ M, J mass attached on the beam we have to consider Rigid body contribute to modal mass (while here NO modal mass effect)

being $W_k(x,t) = \underbrace{\phi_k(x)^T}_{\text{mode of vibration}} \underline{q}(t)$

$$\phi_k^{(i)}(x) = [A_k \cos(\delta_i x) + B_k \sin(\delta_i x) + C_k \cosh(\delta_i x) + D_k \sinh(\delta_i x)]$$

Uncoupled
nat freq there
is a value δ_i
and vibration
mode

$$\left(\begin{matrix} \forall \text{ mat} \\ \text{freq } \omega_k \end{matrix} \right) \quad \gamma_i^4 = \omega_i^2 \frac{m}{EI}$$

with $\underline{\phi}_k(x) = \begin{bmatrix} \phi_k^{(1)} \\ \phi_k^{(2)} \\ \vdots \\ \phi_k^{(n)} \end{bmatrix} \quad \underline{q}(t) = \begin{bmatrix} q_1(t) \\ q_2(t) \\ \vdots \\ q_n(t) \end{bmatrix}$

$[\bar{k}]$ modal are already computed by software

POT. ENERGY (to compute the terms related to K, K_T on $[\bar{k}]$)

$$V = \frac{1}{2} \left\{ \underbrace{\sum_{k=1}^2 \int_0^L W_k''^T EI W_k'' dx}_{\substack{\text{potent energy of} \\ \text{the two} \\ \text{pieces of beam}}} + \underbrace{W_B^T K \cdot W_B}_{\substack{\text{linear} \\ \text{spring pot} \\ \text{em.}}} + \underbrace{\Theta_c^T K_T \Theta_c}_{\substack{\text{pot energy} \\ \text{due to} \\ \text{torsional} \\ \text{spring}}} \right\}$$

$$W_k^T(x,t) = \underline{q}^T(t) \underline{\phi}_k(x) \quad W_k''(x,t) = (\underline{\phi}_k(x))''^T \underline{q} \quad \uparrow \text{Rewriting it...}$$

$$V = \frac{1}{2} q^T \left\{ \underbrace{\left(\sum_{k=1}^2 \int_0^L \phi_k'' EI (\phi_k'')^T dx \right)}_{\text{SOFTWARE implementation}} + \underbrace{\phi_2(0) K (\phi_2(0))^T}_{\text{to add sm MATLAB SCRIPT}} + \underbrace{\left. \frac{\partial \phi_2}{\partial x} \right|_L K_T \left(\left. \frac{\partial \phi_2}{\partial x} \right|_L \right)^T}_{\text{inside the for loop iterating } \forall i \text{ mode}} \right\} q$$

↓
We write contribution of mode (i) inside the for loop

MATLAB

- We write the matrix [H] or an appropriate function
- then define sm main script the parameter EI, L, m, K, Kt and name func = 'H definition'

↓ implement modes of vibration and f_i

Plotting modal shapes

↓

giving as output W, W', W'' derived respect x

W(i mode, :)

↑
rows collect the displacement of each vibration mode

↑ columns collect the values computed in samples dx of beam → dx = 0.001
sampling point 1mm

↓
one MATRIX

only collecting W₁, W₂ in a single output W as full
so to compute beam bending displacement

W₂(L) = W(2L)

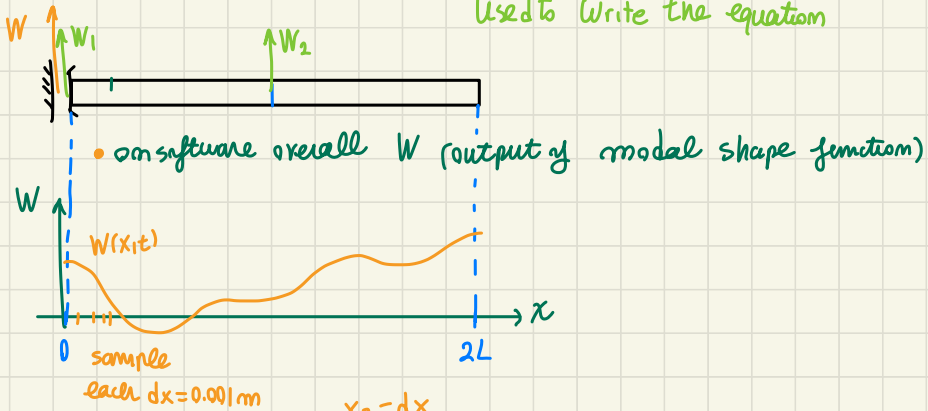
overall bending on BAR 1+2

← from 0 to 2L

W_B = W(L) = central column of W

if interested on $W(L) = W_B$

Used to write the equation



$$W = \begin{bmatrix} W_{mod1}(0) & W_{mod1}(x_2) & \dots & W_{mod1}(L) & \dots & W_{mod1}(2L) \\ W_{mod2}(0) & W_{mod2}(x_2) & \dots & W_{mod2}(L) & \dots & W_{mod2}(2L) \\ \vdots & \vdots & & \vdots & & \vdots \\ W_{modN}(0) & W_{modN}(x_2) & \dots & W_{modN}(L) & \dots & W_{modN}(2L) \end{bmatrix}$$

mod of vibration

we have to find index of this quantity

section B displacement is on the middle of BEAM

sample of function fixed dx

@ $x=L(MIDDLE)$
column index:

→ compute modal stiffness yet done by the code
we use $\rightarrow$ to add K, K_T contribution

$$\begin{aligned} \text{Imd1} &= \text{length}(x)/2 \text{ section B} \\ \text{Imd2} &= \text{length}(x) \text{ section C} \end{aligned} \quad \left\{ \begin{aligned} [K_1] &= \phi_2(0) K (\phi_2(0))^T \\ [K_2] &= \phi_2'(L) K_T (\phi_2'(L))^T \end{aligned} \right\}$$

$$\text{modal_stiff}(\text{imode}) = \text{modal_stiff}(\text{imode}) +$$

$$W(\text{imode}, \text{Imd1})^2 * K + W_p(\text{imode}, \text{Imd2})^2 * K_T \quad \phi_2(0) = W_B \quad \phi_2(L) = W_C$$

inside loop,
as scalar quantity!

$\forall$ imode of vibration we find contribution

ind 1 access $x=L$ displacement (middle of x vector)

$x = m \cdot \text{dom}$ collect all sample points ,

ind 2 access $x=2L$ right end of beam

other parameters remain equal, FRF is already written

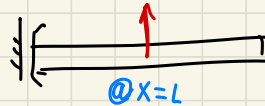
↓

we should compute correctly the modal Lagrangian component of external force F_0

$$Q_m = W(i_{\text{mode}}, \text{ind_input})^* F_0$$

displacement
of section

ind-input = length(x)/2
ind-output = ind-input



applied @ $x=L$

OUTPUT @ $x=L$ of FRF

Where the
force is
applied

done $\forall$ mode of
vibration

q is the modal coordinate of mode of vibration of imode

$$W/W = q^* (W) (i_{\text{mode}}, \text{ind_output})$$

↑
physical
displacement
due to mode i
(imode)

||
 ϕ mode of
vibration

↳ then we sum all

@ end of program @ line 83 → nat freq computed by $\gamma^4 = \frac{\omega^2 m}{EI}$

while, we
can recompute

← from
formula
KNOWN

$$\omega_i = \sqrt{\gamma_i^4 \frac{EI}{m}}$$

matfreq 2 using modal
stiffness and mass

||

matfreq 1 VERY SIMILAR!

$$= \omega_i = \sqrt{\frac{k_{ii}}{m_{ii}}}$$

RESPECT
modal
approach

if wrong computation of
modal parameters they are different!